EXPERIMENT 40

Launch Hadoop 2.x and Test the MapReduce Platform

AIM

To launch the configured Hadoop 2.x single-node cluster and test the MapReduce platform by running a sample word-count job.

REQUIREMENTS

- Hadoop 2.x installed and configured (see Experiment 39)
- HDFS NameNode already formatted
- hduser account with passwordless SSH to localhost
- Sample input text file for the MapReduce word-count job

PROCEDURE

1. Start the HDFS daemons (NameNode, DataNode, Secondary NameNode) using start-dfs.sh.
2. Start the YARN daemons (ResourceManager, NodeManager) using start-yarn.sh.
3. Run the jps command to verify that all required Hadoop daemons are running.
4. Create an input directory in HDFS using the hdfs dfs -mkdir command.
5. Upload a sample text file into the HDFS input directory using hdfs dfs -put.
6. Run the bundled word-count MapReduce example jar against the input directory.
7. Monitor the job progress until the map and reduce phases reach 100 percent completion.
8. View the job output using the hdfs dfs -cat command on the output directory.
9. Stop the daemons using stop-yarn.sh and stop-dfs.sh once testing is complete.

IMPLEMENTATION

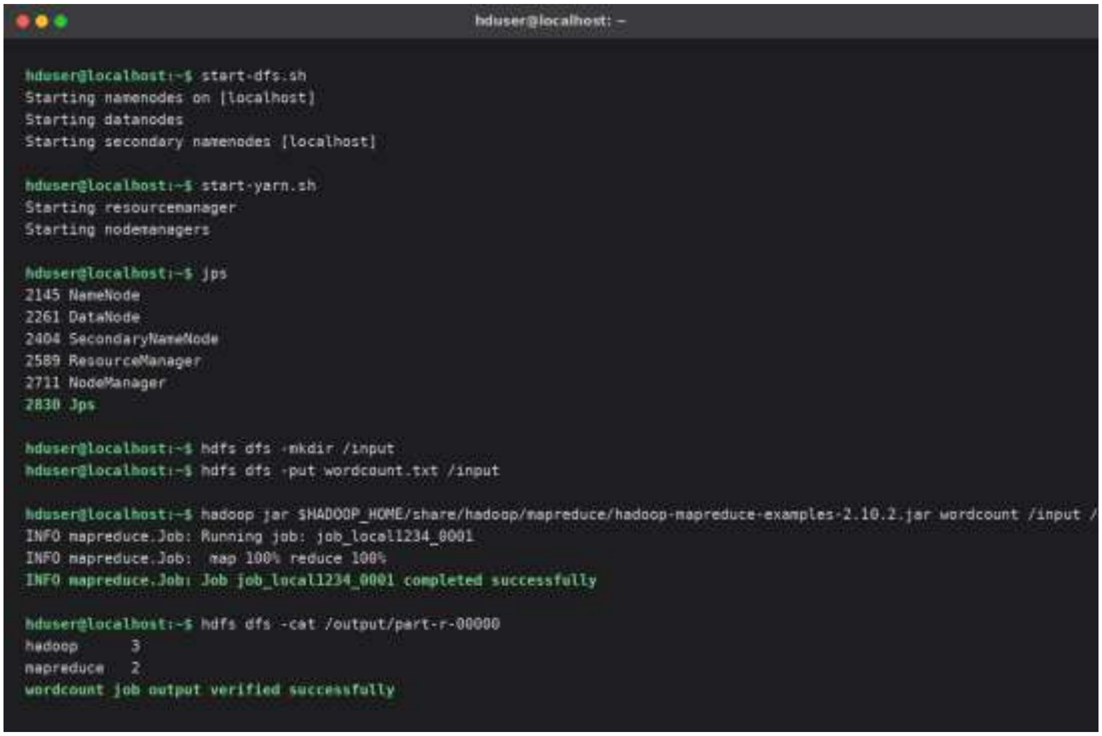

Figure 1: Illustrative terminal view — Hadoop cluster startup and MapReduce word-count test.

RESULT

The Hadoop 2.x cluster was launched successfully with all daemons active, and the sample MapReduce word-count job completed successfully with verified output.